

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 212 — Past Questions & Materials
by Renance

Course: GET 212
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200

This study pack was compiled and restructured by Renance for GET 212 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 4 — GET 212 (Engineer in society)

Level 200 • First Semester • PastQuestion

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QUESTION 3

- (a) Define the principle of relativity. (8 Marks)
- (b) An electric field exerts a force of $8.0 \times 10^{14} \text{ N}$ on a photon moving parallel to it. Determine the magnitudes of momentum and acceleration at the instants when the photon's velocity $v = 0.010c$, $0.90c$ and $0.99c$. (rest mass of photon = $1.67 \times 10^{-27} \text{ kg}$, speed of light $c = 3.0 \times 10^8 \text{ m/s}$). (12 Marks)

(20 Marks)

QUESTION 4

- (a) (i) State Planck's theory of energy quantization (5 marks)
- (ii) State Einstein's photoelectric equation and give the physical meaning of each term in the equation (5 marks)
- (b) An incident X-ray of 0.071 nm is incident on a calcite target. (i) Find the wavelength of the X-ray scattered at an angle of 30° . (ii) Calculate the largest shift that can be expected in this experiment? (10 marks).

(20 Marks)

QUESTION 5

- (a) Define the following terms (i) Florescence of an electron (ii) Binding energy (iii) (4 Marks)
- (b) (i) List three of the emission series observed when electrons in an atom transition from higher energy levels to lower ones. State the electromagnetic waves emitted and the final energy level of the electron for each of the series. (6 marks).
- (ii) An electric magnetic radiation source of power $6 \times 10^{-3} \text{ W}$ emits 1×10^{16} photons per second. The most energetic photoelectron ejected from a metal surface is stopped by a potential difference of 2.2 V . Calculate the work function of the metal. ($M_e = 9.1 \times 10^{-31} \text{ kg}$; $1e = 1.6 \times 10^{-19} \text{ C}$) (10 Marks)

(20 Marks)

Paper 2 of 4 — GET 212 2020-2021 By Akportu Prince PQ

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES. EF FURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

First Semester Examination, 2020/2021 Session

Course Title: Engineering in Society Credit Units: 2

Course Code: GET 212 Time Allowed: 2:00 Hours

Course Code: GET 212

Instruction: Answer Any Four (4) Questions

1. (a) State two obligations of the Engineer to his colleagues.

(b) State three positive impacts of technological advancement in Nigeria

(c) Justify why you should be acquainted with the legal aspect of engineering practice

2. (a) In terms of driving force, what differentiates the Engineer from a quack in project handling?

(b) State the membership grades of the NSE.

(c) List four WAPOR areas of specialization in Civil Engineering

3. (a) Give seven (7) character traits of the Engineer. (6 marks)

(b) State two (2) objectives of the Engineering Regulations Monitoring — § 1.1

(c) Explain the term plagiarism. (4 marks)

4. (a) Define Engineering. (2 marks)

(b) List the members of the engineering family and the basic academic qualification for each cadre

(c) State the professional body for any three divisions of the Nigerian Society of Engineers (NSE).

5. (a) Briefly discuss the fire triangle

(b) State three benefits of optimization in project handling

(c) State three negative impacts of technological advancement in South-south Nigeria

6. (a) Differentiate unsafe act from unsafe condition

(b) With the [killing collapse in mind, state three reasons why | nicians should adhere to their professional codes and standards

AKPORTU PRINCE

Paper 3 of 4 — GET 212 Computer programming session 2018/2019

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester
SESSION: 2018/2019



E CODE: GET 212

COURSE TITLE: Computer
Programming

DATE: 11/05/2019

TIME ALLOWED: 2Hrs

Instruction: Answer FOUR Questions

QUESTION ONE

Write a complete program in C++ to solve the quadratic equation $ax^2 + bx + c = 0$; $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Be sure that the program can handle the special cases; where $a = 0$, where the two roots are equal, and where the discriminant $b^2 - 4ac$ is negative. (17.5 marks)

QUESTION TWO

- Describe the following commands in MATLAB %, length, clc, diary, and clear.
- In tabular form state the difference between script files and function files.
- What is a Computer Language, state and explain the three categories. (17.5 marks)

QUESTION THREE

- In tabular form differentiate between main memory (RAM) and secondary memory (Hard drive).
- State five major operations or functions of a computer.
- State five characteristics of a computers that have made them so powerful and universally useful. (17.5 marks)

QUESTION FOUR

Write a MATLAB function to obtain the roots of the quadratic equation $ax^2 + bx + c = 0$ (17.5marks)

QUESTION FIVE

- Explain with examples the difference between system software and application software of a computer.
- State and explain the three main parts of a central processing unit (CPU) of a computer.
- State the three types of secondary memory of a computer. (17.5 marks)

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QUESTION SIX

For the circuit of figure Q6 shown below, write a MATLAB script to find the nodal voltages V_1 , V_2 and V_3 . (17.5 marks)

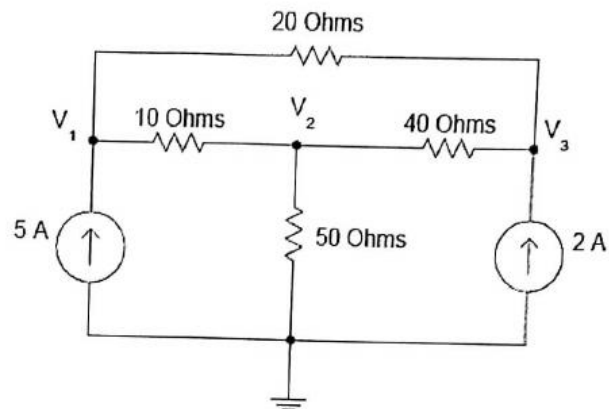


Figure Q6

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester
SESSION: 2017/2018



COURSE CODE: GET 222
COURSE TITLE: Information
Technology in Engineering

DATE: 17/09/2018

TIME ALLOWED: 2Hrs

Instruction: Answer Questions ONE and Any other THREE

QUESTION ONE

(1a) In the industrial automation field, analog signals are still distributed using the venerable 4mA to 20mA current loop. In this technique, the output from any transducer is conditioned by means of some electronics to generate a current of 4mA at the bottom of the scale and vary continuously up to 20mA at the top of the scale. The block diagram in Figure 1 describes just such an application where a 0Vdc to 10Vdc input signal is translated to 4-20mA output. The transfer function is $I_{out} = ((V_{in}/V_{fullscale}) * 16) + 4$. The current, I_{out} , is measured in milliamps. Generate the table for the relationship between the input voltage and the output current using traditional spreadsheet. (4Mks)

(1b) from the table in (1a) explain how to plot 'Scatter with Straight lines and Markers' chart to analyze the linearity of the system using Microsoft excel 2016. (5Mks)

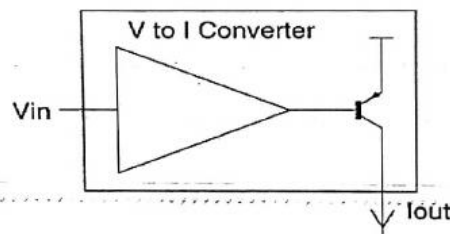


Figure 1 0Vdc to +10Vdc is converted to 4-20mA through this module

(1c) Explain Database Schema. (5Mks)

(1d) State the functions performed by the following shortcut keys: (i) CTRL+Shift+G (ii) CTRL+Shift+I (iii) CTRL+Alt+K (iv) CTRL+Shift+C (v) CTRL+Shift+U (vi) CTRL+Alt+W (vii) Alt+Spacebar (viii) Shift+WindowsKey+M (ix) Alt+Shift+F1 (x) Alt+Shift+T (10Mks)

(1e) Name and explain the parts labelled in Figure 2. (10Mks)

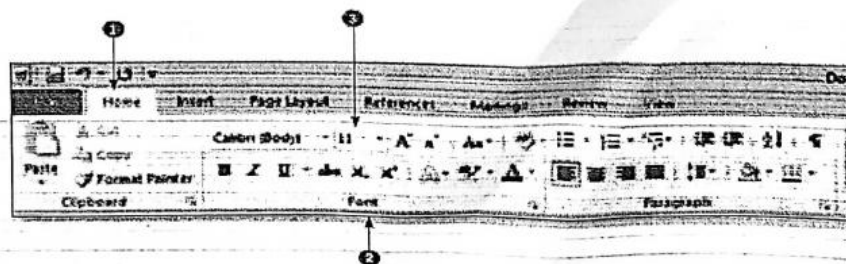


Figure 2

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QUESTION TWO

(2a) (i) What is a Microsoft PowerPoint icon? (ii) List four different elements you can use or add while creating a Microsoft PowerPoint presentation. (3Mks)

(2b) (i) Differentiate between transition and animation in Microsoft PowerPoint. (ii) List the major views in Microsoft PowerPoint. (4Mks)

(2c) Outline the five basic steps you can take to create a Microsoft PowerPoint presentation on the Role of Information Technology in Combating Terrorism. (5Mks)

QUESTION THREE

(3a) A university is preparing admission letters for its over 5000 newly admitted students. The letters should contain each student's address, phone number and name. Outline three basic steps in doing this in MS-Word (2016). (3Mks)

(3b) How do you write this expression, $\int_1^2 8x^2 + \frac{1}{2}x \, dx$ using MS Word (2016)? (4Mks)

(3c) List any five (5) items found in the File tab in MS Word (2016) (5Mks)

QUESTION FOUR

(4a) Define database management system? (2Mks)

(4b) List any eight applications of DBMS. (5Mks)

(4c) What are the advantages of using a DBMS? (5Mks)

QUESTION FIVE

Represent the student data below in a MS Excel sheet and then write one instance of formulae and or functions to extract the following information:

(5a) Complete the columns for total score (1 Mk)

(5b) Complete the column for grades as follows (i) 70 and above = distinction, (ii) 60 to 69 = merit, (iii) 40 to 59 = pass, (iv) less than 40 = fail. (2 Mks)

(5c) Compute number of students with score equal to and greater than 50. (1 Mk)

(5d) Compute number of students who scored between 45 and 65. (2 Mks)

(5e) Compute the mean score of all the students. (1 Mk)

(5f) Write a formula to identify the highest score. (1 Mk)

(5g) Write a formula to identify the lowest score. (1 Mk)

(5h) Write a formula to assign positions to the students according to their scores. Where the first position goes to the person with the highest score (1 Mk)

(5i) Draw and label an actual bar chart to show the distribution of student scores according to the categories in (b) above (1 Mk)

(5j) Draw and label an actual pie chart to show the distribution of student scores according to the categories in (b) above (1 Mk)

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SOURCES: EFFURUN COLLEGE OF TECHNOLOGY

Table 1

	C	D	E	F	G	H	I
1	Mat No	Name	Test (30%)	Exam (70%)	Total (100%)	Grade	Positions
2	1	Alex John	12	25			
3	2	Betty Grace	5	52			
4	3	Ufoma Elohor	16	34			
5	4	Christen James	17	65			
6	5	Christy Peace	10	33			
7	6	Akpos Obukowho	14	27			
8	7	James Bond	20	46			
9	8	Ajiri Uyouyo	3	16			
10	9	Joseph Chukwudi	9	52			
11	10	Tracy John	27	60			

QUESTION SIX

State the functions of the parts labelled in Figure 3

(12Mks)

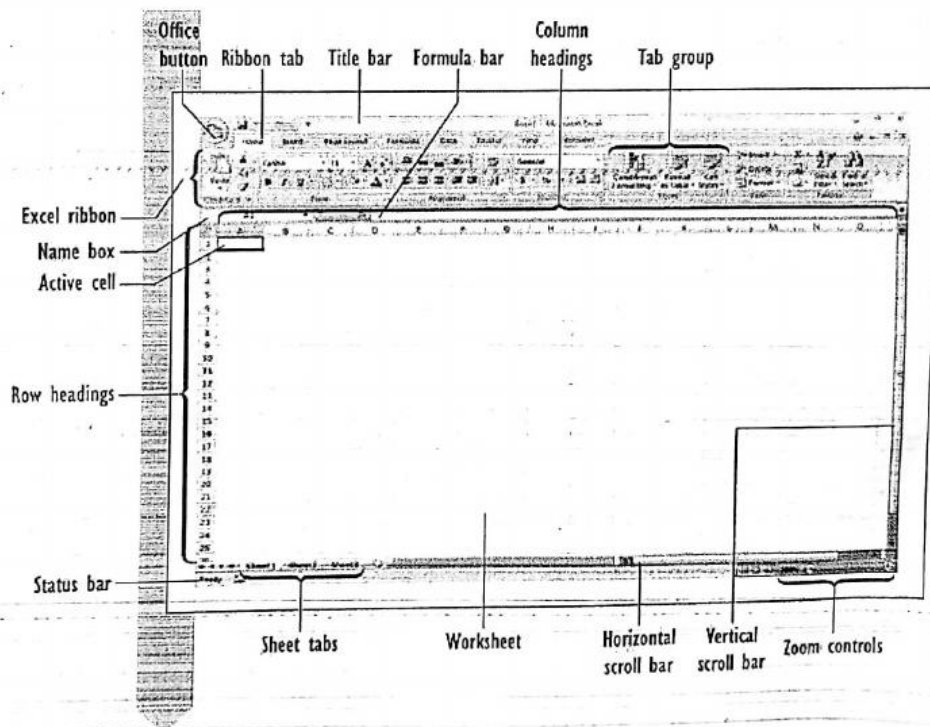


Figure 3

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2018/2019



E CODE: GET 212

COURSE TITLE: Computer Programming

DATE: 11/05/2019

TIME ALLOWED: 2Hrs

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QUESTION TWO

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QUESTION FIVE

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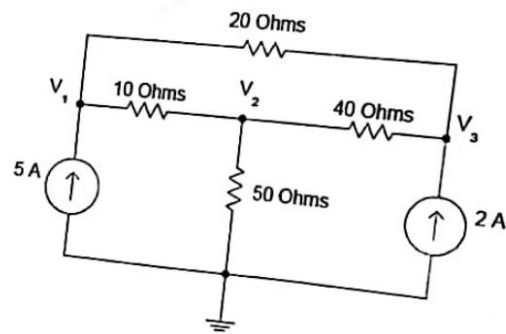


Figure Q6

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Paper 4 of 4 — GET 212 PAST QUESTION

Level 200 • First Semester • PastQuestion

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COLLEGE OF ENGINEERING AND TECHNOLOGY

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3. (a) Give seven (7) character traits of the Engineer. (6 marks)

(b) State two (2) objectives of the Engineering Regulations Monitoring — § 1(1) of the Engineering Regulations Monitoring Act, 2011

(c) Explain the term plagiarism. (4 marks)

4. (a) Define Engineering. (2 marks)

(b) List the members of the engineering family and the basic academic qualification for each cadre

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(c) State three negative impacts of technological advancement in South-south Nigeria

7. (a) Differentiate unsafe act from unsafe condition

(b) With the [killing] collapse in mind, state three reasons why engineers should adhere to their professional codes and standards

(c) State three negative impacts of technological advancement in South-south Nigeria

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